CS 506: Quantum Computing
Scribe Notes
Period Finding via QFT

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1 Overview

In this lecture, we analyze the **core quantum subroutine of Shor's algorithm**: recovering the period r of the function

$$f(x) = a^x \mod N.$$

We build on the measurement state after applying U_f, and analyze how the **Quantum Fourier Transform (QFT)** reveals information about r.

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2 Post-Measurement State

After applying Hadamard gates and the U_f oracle, and measuring the second register, the system collapses to:

$$|\phi\rangle = \frac{1}{\sqrt{Q}} \sum_{k=0}^{Q-1} |x_0 + kr\rangle$$

where:

- r is the period
- x_0 is some offset
- $Q \approx \dfrac{2^m}{r}$

More precisely:

$$Q = \left\lfloor \frac{2^m - x_0 - 1}{r} \right\rfloor + 1$$

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3 Applying the Quantum Fourier Transform

We now apply the QFT over $\mathbb{Z}_{2^m}$:

$$\mathrm{QFT}_{2^m} |x\rangle = \frac{1}{\sqrt{2^m}} \sum_{y=0}^{2^m-1} e^{2\pi i x y/2^m} |y\rangle$$

Applying this to $|\phi\rangle$:

$$\mathrm{QFT}_{2^m} |\phi\rangle = \frac{1}{\sqrt{Q}} \sum_{k=0}^{Q-1} \mathrm{QFT}_{2^m} |x_0 + kr\rangle = \frac{1}{\sqrt{Q}} \sum_{k=0}^{Q-1} \left(\frac{1}{\sqrt{2^m}} \sum_{y=0}^{2^m-1} e^{2\pi i (x_0+kr) y/2^m} |y\rangle \right)$$

Rearranging:

$$= \frac{1}{\sqrt{2^m Q}} \sum_{y=0}^{2^m-1} \left(\sum_{k=0}^{Q-1} e^{2\pi i (x_0+kr) y/2^m} \right) |y\rangle$$

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4 Measurement Probability

The probability of observing $y = y_0$ is:

$$\Pr[y = y_0] = \left| \frac{1}{\sqrt{2^m Q}} \sum_{k=0}^{Q-1} e^{2\pi i (x_0+kr) y_0/2^m} \right|^2$$

Factoring out the x_0 term:

$$= \left| \frac{1}{\sqrt{2^m Q}} e^{2\pi i x_0 y_0/2^m} \sum_{k=0}^{Q-1} e^{2\pi i k r y_0/2^m} \right|^2$$

Since $|e^{i\theta}|^2 = \cos^2\theta + \sin^2\theta = 1$, this simplifies to:

$$\Pr[y = y_0] = \frac{1}{2^m Q} \left| \sum_{k=0}^{Q-1} e^{2\pi i k r y_0/2^m} \right|^2$$

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5 Geometric Series Evaluation

Let:

$$\theta = \frac{2\pi r y_0}{2^m}$$

The sum $\sum_{k=0}^{Q-1} e^{ik\theta}$ is a geometric series with first term $a = 1$ and ratio $s = e^{i\theta}$:

$$\sum_{k=0}^{Q-1} e^{ik\theta} = \frac{1 - e^{iQ\theta}}{1 - e^{i\theta}}$$

Thus:

$$\Pr[y = y_0] = \frac{1}{2^m Q} \left| \frac{1 - e^{iQ\theta}}{1 - e^{i\theta}} \right|^2$$

Using Euler's formula $e^{ix} = \cos x + i \sin x$, we have $e^{ix} - e^{-ix} = 2i \sin x$, so we can write:

$$|1 - e^{i\alpha}|^2 = \left| e^{i\alpha/2}(e^{i\alpha/2} - e^{-i\alpha/2}) \right|^2 = |2i \sin(\alpha/2)|^2 = 4 \sin^2(\alpha/2)$$

Therefore:

$$\Pr[y = y_0] = \frac{1}{2^m Q} \frac{\sin^2(Q\theta/2)}{\sin^2(\theta/2)}$$

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6 Special Case: r Divides 2^m

If $\frac{2^m}{r} \in \mathbb{Z}$, then $\frac{2^m}{r}$ is an integer and $Q = \frac{2^m}{r}$. The measurement probability becomes:

$$\Pr[y_0 = kQ] = \frac{1}{r}, \quad k \in \{0, 1, \ldots, r-1\}$$

and

$$\Pr[y_0 \neq kQ] = 0$$

So the QFT output is **uniformly distributed** over the r exact multiples of $Q = \frac{2^m}{r}$.

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7 General Case: r Does Not Divide 2^m

Now suppose $r \nmid 2^m$. Then y_0 need not be an exact multiple of $\frac{2^m}{r}$. We write:

$$y_0 = k \cdot \frac{2^m}{r} + \delta_k, \quad |\delta_k| < \frac{1}{2}$$

The integer k is drawn uniformly from $\{0, 1, \ldots, r-1\}$, and δ_k is the fractional offset. We now compute $\Pr\left[y = k \cdot \frac{2^m}{r} + \delta_k\right]$.

Example: With $2^m = 2^{10} = 1024$ and $r = 7$:

$$\frac{2^{10}}{7} = \frac{1024}{7} \approx 146.28$$

So observed values cluster near $146, 147, 292, 293, \ldots$

Probability Calculation for General y_0

Starting from:

$$\Pr\left[y = k \cdot \frac{2^m}{r} + \delta_k\right] = \frac{1}{2^m Q} \left| \sum_{z=0}^{Q-1} e^{2\pi i z r y_0 / 2^m} \right|^2$$

Substituting $y_0 = k \cdot \frac{2^m}{r} + \delta_k$:

$$e^{2\pi izry_0/2^m} = e^{2\pi izr\left(k\cdot\frac{2^m}{r}+\delta_k\right)/2^m} = e^{2\pi izk}\cdot e^{2\pi izr\delta_k/2^m}$$

Since $e^{2\pi izk} = 1$ for integer zk (as $e^{\text{integer}\cdot 2\pi i} = 1$), this simplifies to:

$$\Pr\left[y = k\cdot\frac{2^m}{r} + \delta_k\right] = \frac{1}{2^m Q}\left|\sum_{z=0}^{Q-1} e^{2\pi izr\delta_k/2^m}\right|^2$$

This is again a geometric series with $a = 1$ and $s = e^{2\pi ir\delta_k/2^m}$. Applying the closed-form formula and simplifying using the sine identity (as in Section 5):

$$= \frac{1}{2^m Q}\cdot\frac{\sin^2(Q\cdot\pi r\delta_k/2^m)}{\sin^2(\pi r\delta_k/2^m)}$$

Simplification Using θ_1 and θ_2

Setting $\theta_1 = Q\cdot\dfrac{2\pi r\delta_k}{2^m}$ and $\theta_2 = \dfrac{2\pi r\delta_k}{2^m}$, the expression becomes:

$$= \frac{1}{2^m Q}\cdot\left|e^{i(\theta_1-\theta_2)/2}\right|^2\cdot\frac{\sin^2(\theta_1/2)}{\sin^2(\theta_2/2)} = \frac{1}{2^m Q}\cdot\frac{\sin^2(\pi r\delta_k Q/2^m)}{\sin^2(\pi r\delta_k/2^m)}$$

(The complex exponential factor has modulus 1 and drops out.)

Lower Bound on Probability

Using the approximation $Q \approx \dfrac{2^m}{r}$ (so $\dfrac{Qr}{2^m} \approx 1$) and the bound $|\delta_k| < \dfrac{1}{2}$, we get $|\pi r\delta_k/2^m| < \dfrac{\pi r}{2\cdot 2^m}$, which is small. Using $|\sin x| \geq \dfrac{2|x|}{\pi}$ for $|x| \leq \dfrac{\pi}{2}$:

$$\Pr\left[y = k\cdot\frac{2^m}{r} + \delta_k\right] \geq \frac{1}{2^m Q}\cdot\frac{4}{\pi^2} \gtrsim \frac{1}{2N^2}$$

where $N = 2^m$. This confirms that each near-multiple of $\dfrac{2^m}{r}$ has non-negligible measurement probability.

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8 Recovering the Period r

From measurement we observe y_0 satisfying:

$$\frac{y_0}{2^m} \approx \frac{k}{r}$$

We apply the **continued fractions algorithm** to the rational approximation $\dfrac{y_0}{2^m}$ to recover the fraction $\dfrac{k}{r}$ in lowest terms, and then extract r.

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9 Multiple Runs Improve Success Probability

Two-Run Analysis

Suppose we run the algorithm twice, obtaining $y_{0,1}$ and $y_{0,2}$:

$$\frac{y_{0,1}}{2^m} = \frac{k_1}{r} = \frac{c_1}{r_1}, \quad \gcd(c_1, r_1) = 1$$

$$\frac{y_{0,2}}{2^m} = \frac{k_2}{r} = \frac{c_2}{r_2}, \quad \gcd(c_2, r_2) = 1$$

After canceling common factors, we get r_1 and r_2 as divisors of r. Then:

$$r = \operatorname{lcm}(r_1, r_2)$$

with probability:

$$\Pr[r = \operatorname{lcm}(r_1, r_2)] \geq \frac{6}{\pi^2}$$

The GCD Problem and Repetition

The issue is that k and r may share common factors ($\gcd(k, r) > 1$), causing the continued fractions algorithm to return a proper divisor of r rather than r itself.

Key fact: k is chosen uniformly from $\{1, 2, \ldots, r-1\}$ (for $r \geq 19$), and:

$$\Pr[\gcd(k, r) = 1] > \frac{1}{\log \log r}$$

Amplification via Repetition

Run the algorithm $a \cdot \log \log r \leq a \cdot \log \log N$ times (for a constant a). Then:

$$\Pr[\text{at least one run produces } \gcd[k_i, r] = 1] > 1 - \left(1 - \frac{1}{\log \log r}\right)^{2 \log \log r} > 1 - \frac{1}{e^2}$$

So after $O(\log \log N)$ repetitions, the probability of successfully recovering r exceeds $1 - e^{-2}$.

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10 Key Takeaways

- The QFT transforms periodic structure into measurable frequency peaks
- Measurement outcomes are near (or exactly at) multiples of $\frac{2^m}{r}$
- Continued fractions recover $\frac{k}{r}$ from the observed $\frac{y_0}{2^m}$
- Repetition ($O(\log \log N)$ runs) boosts the probability of recovering r to $> 1 - e^{-2}$
- The probability of each near-multiple is $\Omega(1/N^2)$, ensuring non-negligible signal

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11 Connection to Shor's Algorithm

This procedure is the **quantum core** of Shor's algorithm:

1. Reduce factoring N to finding the order r of a mod N
2. Use the QFT subroutine above to extract r quantumly
3. Use classical post-processing (GCD computations) to compute factors of N